

TASK 1

BLUE TEAM REPORT - HAWKEYE LAB

1. Introduction

This report presents a Blue Team analysis of the HawkEye Lab, focusing on detecting, investigating and responding to a simulated cyberattack on the victim system using network traffic data. The goal is to identify indicators of compromise, analyse attacker behaviour and propose mitigation strategies.

2. Objective

- Reconstruct attack timeline
- Analyse attacker behaviour and goal
- Identify compromised data and system
- Mitigate attack and recover system to original state.

3. Tools Used

- Wireshark – analyses network traffic
- CyberChef – decodes Base64 encoded data
- Windows PowerShell – retrieves the file hash for malware verification
- IPlocation.net – provides geolocation of an IP Address.

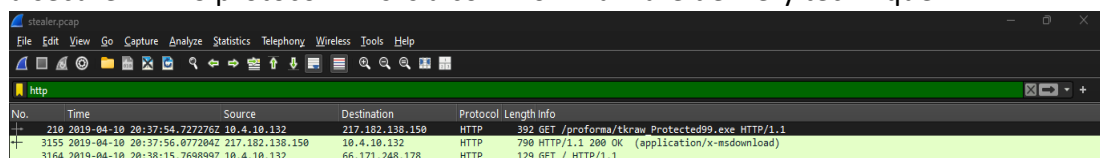
4. Incident Overview

An accountant at your organization received an email regarding an invoice with a download link. Suspicious network traffic was observed shortly after opening the email. As a SOC analyst, investigate the network trace and analyse exfiltration attempts.

5. Detection Analysis

5.1. HTTP Detection

An executable file is detected to be downloaded over **unencrypted HTTP** rather than a secure HTTPS protocol. This is a common malware delivery technique.



No.	Time	Source	Destination	Protocol	Length	Info
210	2019-04-10 20:32:54.7272762	10.4.10.132	217.182.138.150	HTTP	592	GET /proforma/tknew_Protected99.exe HTTP/1.1
3155	2019-04-10 20:37:56.0772042	217.182.138.150	10.4.10.132	HTTP	790	HTTP/1.1 200 OK (application/x-msdownload)
3164	2019-04-10 20:38:15.7698992	10.4.10.132	66.171.248.178	HTTP	129	GET / HTTP/1.1

Figure 5.1.

5.2. DNS Detection

DNS queries sent to unusual domain, **proforma-invoices.com** that looks suspicious indicating command-and-control (C2C) communication.

```

Hypertext Transfer Protocol
  GET /proforma/tkraw_Protected99.exe HTTP/1.1\r\n
  Accept: */*\r\n
  Accept-Encoding: gzip, deflate\r\n
  User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0;
  Host: proforma-invoices.com\r\n
  Connection: Keep-Alive\r\n
  \r\n
  [Response in frame: 3155]
  [Full request URI: http://proforma-invoices.com/proforma/tkraw_Protected99.exe]

```

Figure 5.2.

5.3. Malware Detection

The presence of a suspicious .exe file, **tkraw_Protected99.exe** indicates existence of malware.

Wireshark · Export · HTTP object list

Text Filter: Content Type: All Content-Types

Packet	Hostname	Content Type	Size	Filename
3155	proforma-invoices.com	application/x-msdownload	2025 kB	tkraw_Protected99.exe
3166	bot.whatismyipaddress.com	text/html	14 bytes	\
3297	bot.whatismyipaddress.com	text/html	14 bytes	\
3384	bot.whatismyipaddress.com	text/html	14 bytes	\
3469	bot.whatismyipaddress.com	text/html	14 bytes	\
3584	bot.whatismyipaddress.com	text/html	14 bytes	\
3839	bot.whatismyipaddress.com	text/html	14 bytes	\
3917	bot.whatismyipaddress.com	text/html	14 bytes	\

Figure 5.3.

5.4. Data Exfiltration Detection

Unusual **SMTP** traffic can be detected by monitoring outgoing network traffic involving sharing sensitive data and abnormal patterns of frequent data transfer every **10 minutes**.

stealer.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

smtp

No.	Time	Source	Destination	Protocol	Length	Info
3194	2019-04-10 20:38:16.7122512	10.4.10.132	23.229.162.69	SMTP	410	C: DATA fragment, 356 bytes
3196	2019-04-10 20:38:16.7123592	10.4.10.132	23.229.162.69	SMTP	1076	C: DATA fragment, 1022 bytes
3198	2019-04-10 20:38:16.7124582	10.4.10.132	23.229.162.69	SMTP	1078	C: DATA fragment, 1024 bytes
3200	2019-04-10 20:38:16.7125162	10.4.10.132	23.229.162.69	SMTP	198	C: DATA fragment, 144 bytes
3202	2019-04-10 20:38:16.7125982	10.4.10.132	23.229.162.69	SMTP	56	C: DATA fragment, 2 bytes
3204	2019-04-10 20:38:16.7126612	10.4.10.132	23.229.162.69	SMTP/I...	59	from: sales.del@macwinlogistics.in, subject: =?utf-8?B?5GF3a0V5ZSBLZXI1b2dnZXIgL585ZWJvcw==?d4jk
3206	2019-04-10 20:38:16.8537042	23.229.162.69	10.4.10.132	SMTP	82	S: 250 OK id=1b2j6-0066e9-AF
3253	2019-04-10 20:39:56.1116822	23.229.162.69	10.4.10.132	SMTP	121	S: 421 p3plcpn10413.prod.phx3.secureserver.net lost input connection
3306	2019-04-10 20:48:20.6464022	23.229.162.69	10.4.10.132	SMTP	251	S: 220-p3plcpn10413.prod.phx3.secureserver.net ESMTP Exim 4.91 #1 Wed, 10 Apr 2019 13:48:20 -0
3307	2019-04-10 20:48:20.6467322	23.229.162.69	10.4.10.132	SMTP	76	C: EHLO Beijing-5cd1-PC
3309	2019-04-10 20:48:20.7158252	23.229.162.69	10.4.10.132	SMTP	261	S: 250-p3plcpn10413.prod.phx3.secureserver.net Hello Beijing-5cd1-PC [173.66.146.112] SIZE 5
3310	2019-04-10 20:48:20.7152592	10.4.10.132	23.229.162.69	SMTP	187	C: AUTH login User: c2FsZXMuZGVsQ6l1Y3dpbmV212ldG1jcy5pbG==
3312	2019-04-10 20:48:20.7823632	23.229.162.69	10.4.10.132	SMTP	72	S: 334 UGfc3dvcvQ6
3313	2019-04-10 20:48:20.7825992	10.4.10.132	23.229.162.69	SMTP	68	C: Pass: U2FsZXMuZGVsQ6l1Y3dpbmV212ldG1jcy5pbG==
3315	2019-04-10 20:48:20.8504212	23.229.162.69	10.4.10.132	SMTP	84	S: 235 Authentication succeeded
3316	2019-04-10 20:48:20.8506162	10.4.10.132	23.229.162.69	SMTP	96	C: MAIL FROM:<sales.del@macwinlogistics.in>
3318	2019-04-10 20:48:20.9146212	23.229.162.69	10.4.10.132	SMTP	62	S: 250 OK

Frame 3179: Packet, 251 bytes on wire (2008 bits), 251 bytes captured (2088 bits) on 0

Ethernet II, Src: Netgear-bd9193f1 (28:e5:12:b6:93:f1), Dst: Hewlett-Packard-1c:47:ae (08:00:02:1c:47:ae)

Internet Protocol Version 4, Src: 23.229.162.69, Dst: 10.4.10.132

Transmission Control Protocol, Src Port: 587, Dst Port: 49206, Seq: 1, Ack: 1, Len: 197

Simple Mail Transfer Protocol

0000 00 02 1c 47 ae 20 e5 2a b6 93 f1 08 00 45 00 ...G...E-
 0010 00 ed 96 c0 00 00 00 06 64 98 17 e5 a2 45 0a 04 ...E..
 0020 0a 04 02 4b c0 36 24 96 2e 4e 07 30 14 f7 50 18 ...K 65 .H 0 .P..
 0030 fa f0 89 fd 00 00 32 32 30 2d 70 33 70 6c 63 70 ...22 0-p3plcp
 0040 6e 6c 30 34 31 33 2e 70 72 6f 64 2e 70 68 78 33 n10413.p rod.phx3
 0050 2e 73 65 63 75 72 65 73 65 72 76 65 72 2e 6e 65 .secureserver.ne
 0060 74 20 45 53 4a 54 50 20 45 78 69 6d 20 34 2e 39 t ESMTP Exim 4.9
 0070 31 20 23 31 20 57 65 64 2c 20 31 30 20 41 70 72 1 #1 Wed , 10 Apr
 0080 20 32 30 31 39 20 31 33 3a 33 38 3a 31 35 20 2d 2019 13 :38:15 -
 0090 30 37 30 30 20 0d 0a 32 32 30 2d 57 65 20 64 6f 0700 . 2 20-We do
 00a0 20 6e 6f 74 20 61 75 74 68 6f 72 69 7a 65 20 74 not aut horize t
 00b0 68 65 20 75 73 65 20 6f 66 20 74 68 69 73 20 73 he use o f this s
 00c0 79 73 74 65 6d 20 74 6f 20 72 61 6f 20 72 61 6e 72 70 6f ystem to transpo
 00d0 72 74 20 75 6e 75 6f 6c 69 69 74 65 64 2c 20 rt unsol icited,
 00e0 0d 0a 32 32 30 20 61 6e 64 2f 6f 72 20 62 75 6c .220 an d/or bul
 00f0 6b 20 65 2d 6d 61 69 6c 2e 0d 0a k e-mail ...

Simple Mail Transfer Protocol Protocol

Packets: 4003 · Displayed: 147 (3.7%)

Profile: Default

33°C Mostly sunny

Search

ENG IN

16:32 03-05-2026

Figure 5.4.

5.5 Indicators of Compromise (IoCs)

1. Malicious Domain

proforma-invoices.com was the domain used by the victim system to download a suspicious file.

```
Host: proforma-invoices.com\r\n
Connection: Keep-Alive\r\n
\r\n
[Response in frame: 3155]
[Full request URI: http://proforma-invoices.com/proforma/tkraw_Protected99.exe]
```

Figure 5.5.1.

2. Malicious IP address

217.182.138.150, extracted by investigating the malicious domain, used for communication.

```
Internet Protocol Version 4, Src: 10.4.10.132, Dst: 217.182.138.150
Transmission Control Protocol, Src Port: 49204, Dst Port: 80, Seq: 1, Ack: 1, Len: 338
Hypertext Transfer Protocol
  GET /proforma/tkraw_Protected99.exe HTTP/1.1\r\n
  Accept: */*\r\n
  Accept-Encoding: gzip, deflate\r\n
  User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0; SLCC2;
```

Figure 5.5.2.

3. Victim Machine IP

10.4.10.132, internal system initiating suspicious traffic with large amount of data transfer and malware download.

Ethernet · 7	IPv4 · 12	IPv6	TCP · 48	UDP · 58			
Address ^	Packets	Bytes	Tx Packets	Tx Bytes	Rx Packets	Rx Bytes	Country
10.4.10.2	42	5 kB	0	0 bytes	42	5 kB	
10.4.10.4	513	114 kB	234	46 kB	279	68 kB	
10.4.10.132	4,003	2 MB	1,993	212 kB	2,010	2 MB	
10.4.10.255	30	3 kB	0	0 bytes	30	3 kB	
23.229.162.69	280	39 kB	161	13 kB	119	26 kB	
66.171.248.178	63	5 kB	28	3 kB	35	2 kB	
216.58.193.131	20	8 kB	11	6 kB	9	3 kB	
217.182.138.150	2,947	2 MB	1,576	2 MB	1,371	74 kB	
224.0.0.22	23	1 kB	0	0 bytes	23	1 kB	
224.0.0.252	10	750 bytes	0	0 bytes	10	750 bytes	
239.255.255.250	74	29 kB	0	0 bytes	74	29 kB	
255.255.255.255	1	342 bytes	0	0 bytes	1	342 bytes	

Figure 5.5.3.

4. Malicious File Name

tkraw_Protected99.exe, .exe indicates executable file indicating malware.

```
Hypertext Transfer Protocol
> GET /proforma/tkraw_Protected99.exe HTTP/1.1\r\n
Accept: */*\r\n
Accept-Encoding: gzip, deflate\r\n
User-Agent: Mozilla/4.0 (compatible; MSIE 7.0; Windows NT 6.1; WOW64; Trident/7.0;
```

Figure 5.5.4.

6. Investigation

6.1. Alert Triage

The incident can be considered as **high-alert** as there is malware detected and continuous data exfiltration activity. By correlating data sources and victim activity, false positives can be ruled out.

6.2. Log Analysis

- Network traffic logs were analysed using Wireshark to identify suspicious communication patterns.
- Use of HTTP for communication indicates suspicious data transfer.

6.3. Root Cause Analysis

- The root cause of the attack was identified as the download of the executable file **tkraw_Protected99.exe** from an untrusted source over HTTP. This indicates a possible phishing or a social engineering attack where the victim was tricked into downloading the file.
- The lack of secure HTTPS communication allowed the attack to go undetected.

6.4. Attack Timeline

1. Victim IP – 10.4.10.132 attempts to access a suspicious domain, **proforma invoices.com**.
2. DNS query sent to DNS server- 10.4.10.4, receives **IP address 217.182.138.150**.
3. Victim connects to 217.182.138.150 over unencrypted **HTTP** rather than HTTPS.
4. Suspicious .exe file, **tkraw_Protected99.exe** downloaded by the victim.
5. Victim OS – **Windows NT 6.1**, more vulnerable since old version, indicates exploiting vulnerability.
6. Communication established with malicious IP.
7. Data exfiltration confirmed after observing SMTP traffic which indicates exfiltration through emails.
8. The malware establishes persistence and continues data exfiltration every 10 minutes indicating automated attacker control, thus confirming that the system is compromised.

7. Containment

7.1. Short-term containment

- The infected system was isolated from the network to stop communication with the attacker-controlled system.
- The malicious IP address and domain identified during analysis were blocked at firewall level to prevent further connections to other systems.

- Any possibly compromised user accounts were disabled to avoid potential exploit.

7.2. Long-term containment

- Network segmentation was deployed to prevent lateral movement of an attacker within the system.
- Access permission were reviewed and restricted according the principle of least privilege.

8. Eradication

8.1. Malware Removal

- The identified malware executable file was completely removed from the infected system. System scans were performed to ensure no other malicious files were present.
- Sensitive information and credentials of user accounts that were potentially exposed during the attack, were asked to reset passwords to eradicate any chance of unauthorized access.
- Any changes made by the malware were identified, reversed and restored back to it clean state.

8.2. System hardening

- Unnecessary services were disabled. Security configurations were applied to prevent risk of future attacks.
- Firewall and other endpoint protection rules were strengthened to prevent execution of unauthorized files.

8.3. Vulnerability fixing

- Due to lack of monitoring controls, the attack leveraged used of unsecure HTTP for communication with an untrusted source.
- New security updates and patches were applied to address this.

9. Recovery

9.1. System Restoration

The affected system was restored to its normal state after eradicating the malware. This can be done with the help of clean backups.

9.2. Validation

- Post-recovery validation was done to ensure that there is no malicious content left in the system.
- System functionality was tested to ensure the normal operations were restored without any performance and security issues.

9.3. Monitoring

Enhanced monitoring was implemented to closely monitor for abnormal patterns or any signs of malware existence in DNS queries and communication over the network.

9.4. Business Continuity

After validating the system stability, critical business operations were resumed without any disruptions along with persistent monitoring.

10. Lessons Learned

10.1. Incident Review

The impact of the attack included potential compromise of sensitive information as there was data exfiltration, unauthorized external communication which indicates weak network monitoring and detection.

10.2. Gap Analysis

- There was absence of strict network controls and real-time monitoring which may have stopped the attack in the previous stages causing lesser data compromise.
- Re-tuning the threshold for suspicious DNS queries and outbound SMTP traffic may have helped in identifying the attacker in advance.

10.3. Training and Awareness

- Awareness programs have to be conducted among employees to train them to identify malware, social engineering attacks and verify the sources before sharing sensitive information.
- Additionally, SOC Analysts have to be familiar with latest tools, and deploy anti-malware software to detect and respond to attacks before they spread.

10.4. Improvements

- Enhanced security monitoring
- Employee training and awareness
- Anti-malware solutions
- Updated detection rules

11. Conclusion

This analysis demonstrates how attackers exploit unencrypted communication channels and email-based exfiltration techniques to compromise systems. A strong detection strategy, combined with continuous monitoring and proactive security controls, is essential to prevent such attacks.